

HDOC - Day 29

Menu-Driven Array Operations in C

Question / Aim

Write a C program to perform the following menu-driven operations on a given array: (1) count even, odd, positive, negative and zero elements, (2) search for an element using linear search, (3) merge two arrays, and (4) exit.

Algorithm

1. Start and input the size and elements of the first array.
2. Display the menu and read the user's choice.
3. For choice 1, traverse the array and count even, odd, positive, negative and zero elements.
4. For choice 2, input a key and scan the array from left to right. Display its position if found; otherwise display not found.
5. For choice 3, input the second array and copy both arrays into a merged array.
6. For choice 4, terminate the program. For any other choice, display Invalid choice.
7. Stop.

Prepared Test Case Table

Test	Operation	Input details	Expected output
1	Counting	[-4, 0, 7, 8, -3, 2, 5]	Even=4, Odd=3, Positive=4, Negative=2, Zero=1
2	Linear search - found	[12, 5, 9, 20, 7, 3], key=20	20 found at position 4
3	Linear search - not found	[4, 8, 12, 16, 20], key=7	7 not found
4	Merge arrays	[1,2,3,4] and [5,6,7]	[1,2,3,4,5,6,7]

C Program

```
#include <stdio.h>
#define MAX 100

int main() {
    int a[MAX], b[MAX], merged[2*MAX];
    int n, n2, i, choice, key, found;
    int even, odd, positive, negative, zero;

    printf("Enter number of elements: ");
    scanf("%d", &n);
    printf("Enter %d elements: ", n);
    for(i = 0; i < n; i++) scanf("%d", &a[i]);

    do {
        printf("\n----- MENU ----- \n");
        printf("1. Count even, odd, positive, negative and zero elements\n");
        printf("2. Linear search\n");
        printf("3. Merge two arrays\n");
        printf("4. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);

        switch(choice) {
            case 1:
                even = odd = positive = negative = zero = 0;
                for(i = 0; i < n; i++) {
                    if(a[i] % 2 == 0) even++;
                    if(a[i] > 0) positive++;
                    else if(a[i] < 0) negative++;
                    else zero++;
                }
                printf("Even numbers      : %d\n", even);
                printf("Odd numbers       : %d\n", odd);
                printf("Positive numbers: %d\n", positive);
                printf("Negative numbers: %d\n", negative);
                printf("Zero elements    : %d\n", zero);
                break;

            case 2:
                printf("Enter element to search: ");
                scanf("%d", &key);
                found = -1;
                for(i = 0; i < n; i++)
                    if(a[i] == key) { found = i; break; }
                break;

            case 3:
                printf("Enter number of elements for second array: ");
                scanf("%d", &n2);
                for(i = 0; i < n2; i++) scanf("%d", &b[i]);
                for(i = 0; i < n; i++) merged[i] = a[i];
                for(i = n; i < n+n2; i++) merged[i] = b[i-n];
                printf("Merged array: ");
                for(i = 0; i < n+n2; i++) printf("%d ", merged[i]);
                printf("\n");
                break;

            case 4:
                break;

            default:
                printf("Invalid choice\n");
                break;
        }
    } while(choice != 4);

    return 0;
}
```

```

        if(found != -1)
            printf("Element %d found at position %d\n", key, found + 1);
        else
            printf("Element %d not found in the array\n", key);
        break;

    case 3:
        printf("Enter size of second array: ");
        scanf("%d", &n2);
        printf("Enter %d elements: ", n2);
        for(i = 0; i < n2; i++) scanf("%d", &b[i]);
        for(i = 0; i < n; i++) merged[i] = a[i];
        for(i = 0; i < n2; i++) merged[n+i] = b[i];
        printf("Merged array: ");
        for(i = 0; i < n+n2; i++) printf("%d ", merged[i]);
        printf("\n");
        break;

    case 4:
        printf("Exiting program.\n");
        break;
    default:
        printf("Invalid choice.\n");
    }
} while(choice != 4);
return 0;
}

```

Compilation, Execution & Testing

Test Case 1 - Counting

```

puppie@Anvikas-MacBook-Air Desktop % ./array29
Enter number of elements: 7
Enter 7 elements: -4 0 7 8 -3 2 5

----- MENU -----
1. Count even, odd, positive, negative and zero elements
2. Linear search
3. Merge two arrays
4. Exit
Enter your choice: 1
Even numbers      : 4
Odd numbers       : 3
Positive numbers: 4
Negative numbers: 2
Zero elements     : 1
puppie@Anvikas-MacBook-Air Desktop %

```

Test Case 2 - Linear Search (Found)

```

puppie@Anvikas-MacBook-Air Desktop % ./array29
Enter number of elements: 6
Enter 6 elements: 12 5 9 20 7 3

----- MENU -----
1. Count even, odd, positive, negative and zero elements
2. Linear search
3. Merge two arrays
4. Exit
Enter your choice: 2
Enter element to search: 20
Element 20 found at position 4
puppie@Anvikas-MacBook-Air Desktop %

```

Test Case 3 - Linear Search (Not Found)

```
puppie@Anvikas-MacBook-Air Desktop % ./array29
Enter number of elements: 5
Enter 5 elements: 4 8 12 16 20

----- MENU -----
1. Count even, odd, positive, negative and zero elements
2. Linear search
3. Merge two arrays
4. Exit
Enter your choice: 2
Enter element to search: 7
Element 7 not found in the array
puppie@Anvikas-MacBook-Air Desktop %
```

Test Case 4 - Merge Two Arrays

```
puppie@Anvikas-MacBook-Air Desktop % ./array29
Enter number of elements: 4
Enter 4 elements: 1 2 3 4

----- MENU -----
1. Count even, odd, positive, negative and zero elements
2. Linear search
3. Merge two arrays
4. Exit
Enter your choice: 3
Enter size of second array: 3
Enter 3 elements: 5 6 7
Merged array: 1 2 3 4 5 6 7
puppie@Anvikas-MacBook-Air Desktop %
```

Result

The menu-driven C program for counting array elements, linear search and merging two arrays was successfully implemented and tested using the prepared test cases.